

2SC7692 Shape optimization and drag reduction in aeronautics

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1 Overview

The steady rise in air traffic raises concerns about the sustainability of greenhouse gas emissions. To mitigate this, improving aircraft energy efficiency is essential. This can be achieved by enhancing engine performance or optimizing aerodynamic design, such as reducing aircraft weight. Computational tools have played a crucial role in aeronautics, aiding in the design and optimization of aircraft components. For instance, refining wing shapes can reduce drag and lift, while optimizing internal structures can decrease overall weight, leading to better fuel efficiency.

2 Tools and Technologies

We utilize the following open-source software tools:

- **Finite Element Software:** FEniCSx (Python)
- **Meshing Tools:** Gmsh, MeshMixer, Rhinoceros – Rhino 8
- **Visualization:** VTK (Paraview)

3 Overview of Worksheets

3.1 Worksheet 1

This worksheet served as an introduction to the FEniCS software, particularly for mesh generation, boundary marking and solving the Poisson problem.

1. Numerical Error Measurement for the Poisson Problem

- L_2 error: 5.38×10^{-3}
- H_1 error: 2.18×10^{-1}

2. Convergence rate of the Finite element Scheme

A log plot of the convergences of the scheme for increasing elements is shown below, with corresponding linear lines of best fit, of the form $y = mx$. Note that with quadratic Lagrange elements, the convergence rate of the H_1 norm becomes second order.

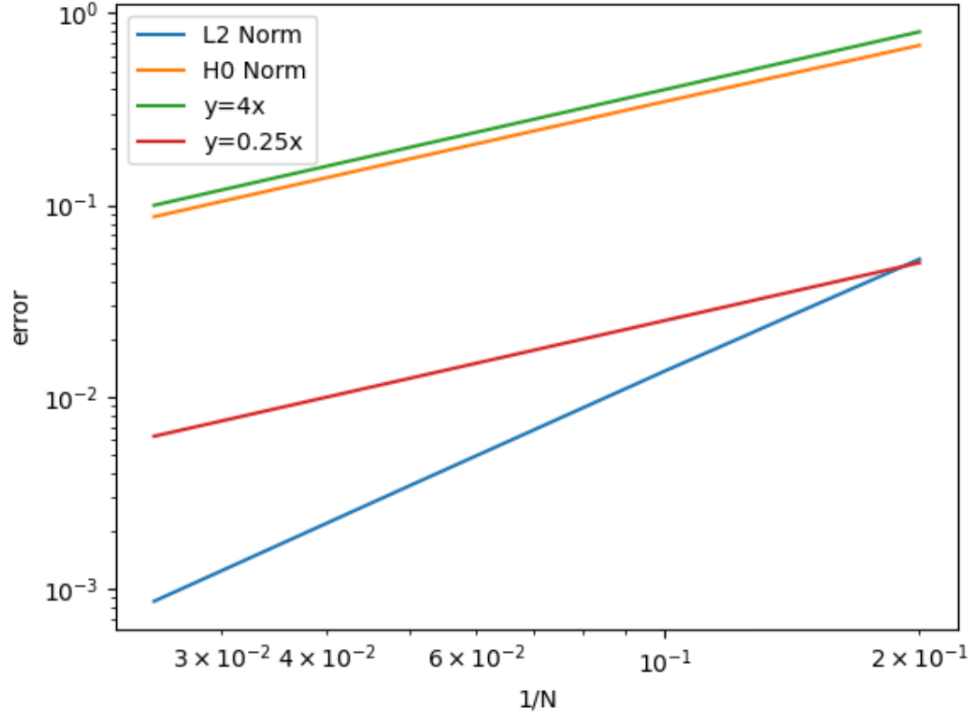


Figure 1: Plot of error against $1/N$.

3. Errors and Computation Time for Different Degrees of Freedom

Number of Degrees of Freedom (N)	L_2 Error	H_1 Error	Computation Time (seconds)
5	5.22×10^{-2}	2.00×10^{-1}	0.0044
10	1.36×10^{-2}	1.00×10^{-1}	0.0045
20	3.45×10^{-3}	5.00×10^{-2}	0.0063
40	8.65×10^{-4}	2.50×10^{-2}	0.0139

Table 1: Errors and Computation Time for Different Degrees of Freedom

Graphically, the increase in computation time with increasing number of degrees of freedom can be seen in figure 2 below.

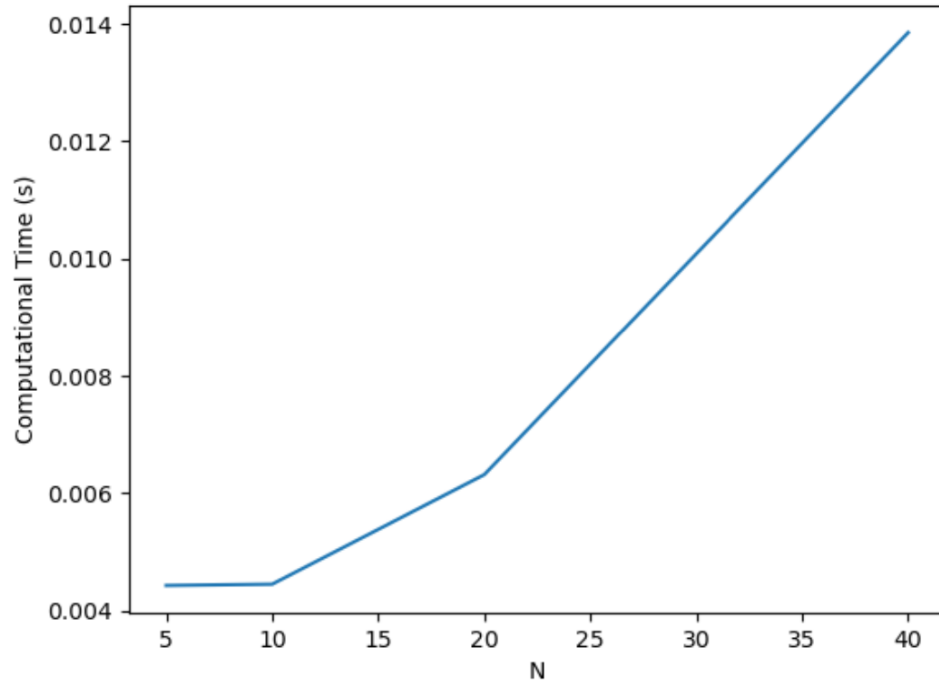


Figure 2: Plot of Computation Time against Number of Degrees of Freedom (N).

4. Mesh Geometry and Poisson Solver with Dirichlet Boundary Conditions

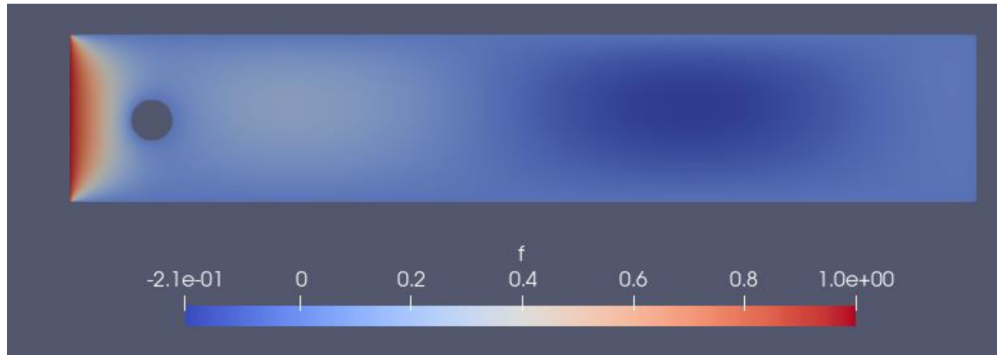


Figure 3: Mesh geometry with Dirichlet boundary conditions $u = 0$ everywhere, except on boundary 1, where $u = 1$.

5. Mixed Poisson Problem with Dirichlet Conditions

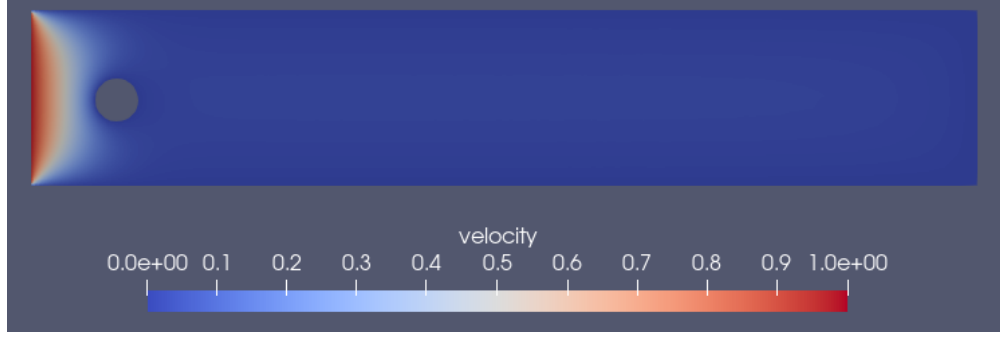


Figure 4: Mixed Poisson Problem with $f_2 = 1$ and same boundary conditions as above

3.2 Worksheet 2

This worksheet broadened the numerical CFD scheme to solve the stokes equations, and applying that to measure drag and lift around objects, with a uniqueness term.

1. Pressure Distribution around the Object

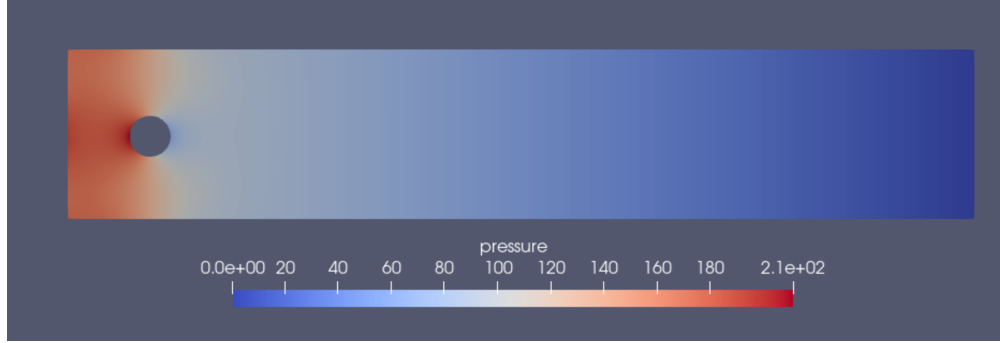


Figure 5: Pressure distribution around the object.

2. Velocity Distribution around the Object

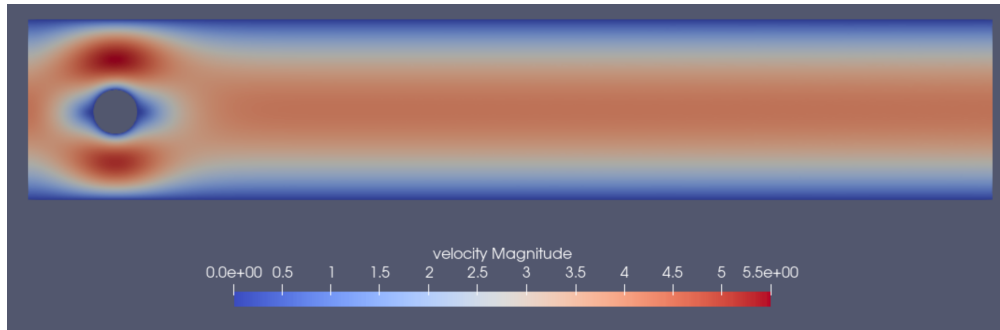


Figure 6: Velocity distribution around the object.

3. Results

- Objective function value J : 136.688
- Lift F_l : -0.585 N
- Drag F_d : -45.136 N

3.3 Worksheet 3

This worksheet revolves around a NACA 2D Airfoil. The main objectives were to create a mesh routine as well as a Computational Fluid Dynamics (CFD) routine, in order to measure the drag and the lift at different angles of attack.

1. Plot of results from 4 different camber values, evaluated from -90° to 90° angle of attacks

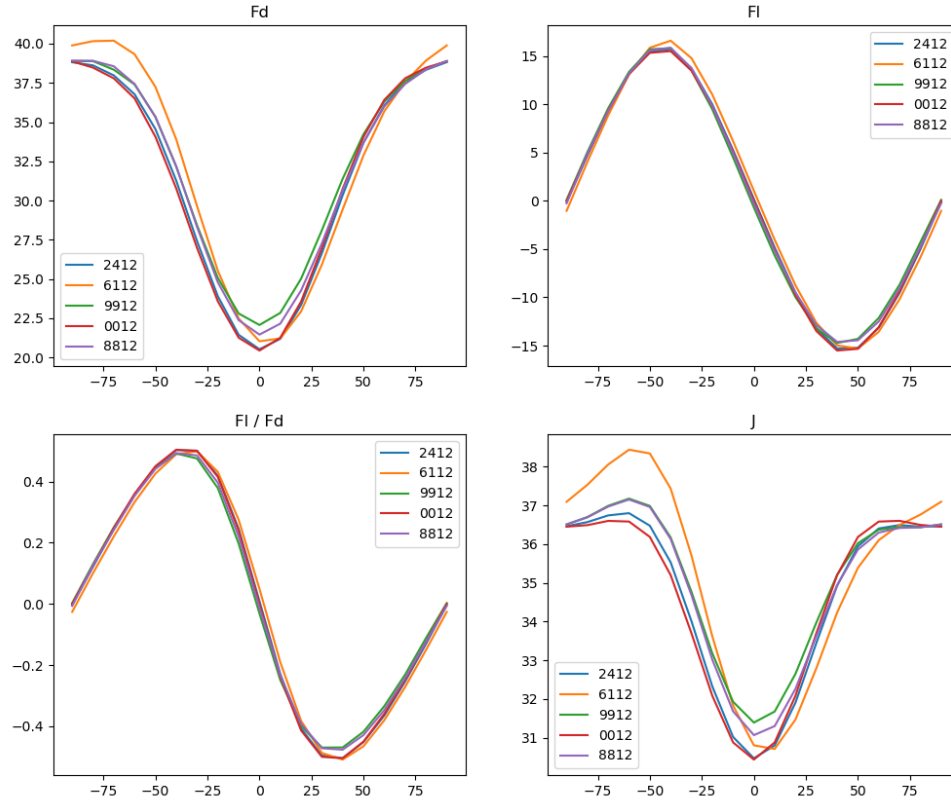


Figure 7: Results of the NACA 2D Airfoil simulation with 4 different cambers

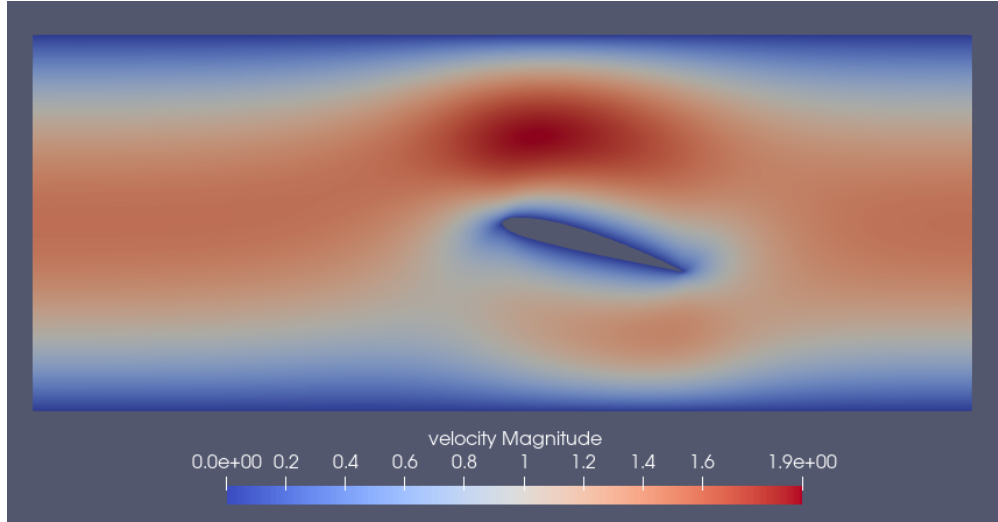


Figure 8: Velocity Field of the NACA 2D Airfoil at 15°

2. Plot of F_D and F_L for 100 values for the pair (α, camber)

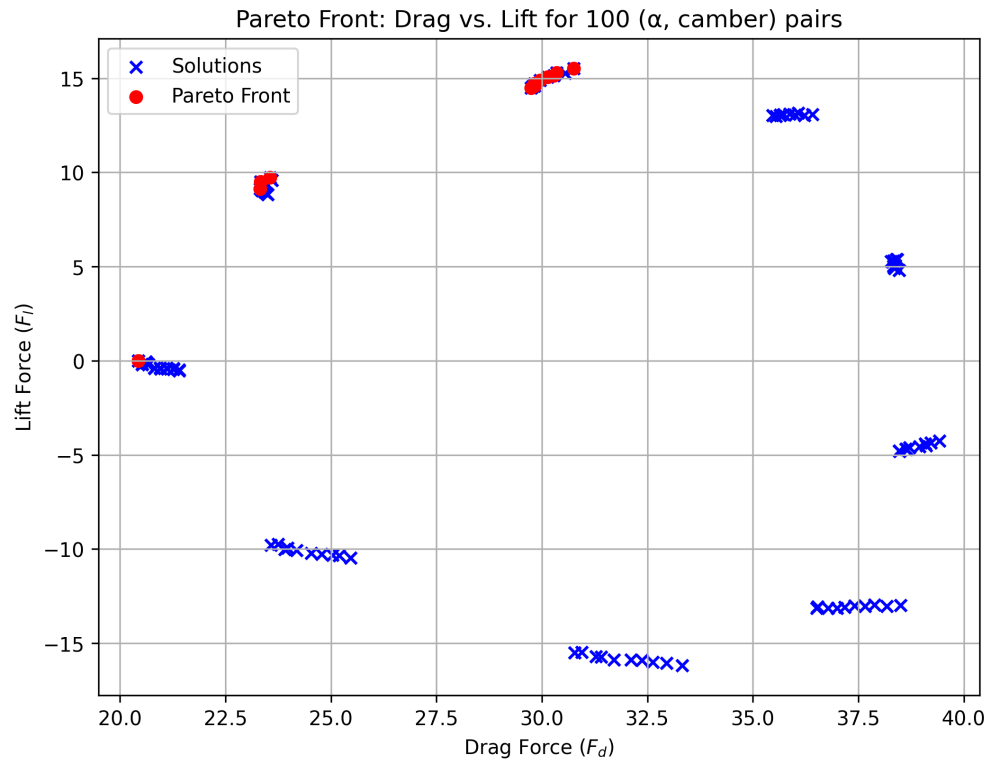


Figure 9: Pareto Front compared to numerical solution of 100 (α , *camber*)

4 Simulations with Boeing 767

We wrote Python scripts that would allow us to simulate the effects of airflow around a model of a 3 dimensional aircraft. Details of the workflow around these simulations are explained below, including meshing, hardware, and an explanation of the results.

4.1 Meshing

4 skin meshes of different refinement levels were created using Rhinoceros 8 Shrink Wrap software. GMSH was then used to generate a volume mesh around the plane, and to angle the bounding box to simulate different angles of attack during flight. Details of these meshes at 0° inclination are tabulated below. Note that the number of nodes and elements varied slightly with changed inclination.

Mesh number	No. of nodes	No. of elements
1 (most coarse)	4852	27413
2	15334	82999
3	24526	128621
4 (least coarse)	53378	282881

Table 2: Mesh details with corresponding number of nodes and elements

4.2 Hardware

3 dimensional CFD is computationally demanding. Simulations were run using the DCE cluster at Centrale-Supélec, using the fenicsx linear problem solving tools run in parallel across multiple CPU processors. In the DCE cluster, the kyle[01-68] nodes were accessed, which have 32 Intel Xeon Silver 4110 @ 2.1GHz, CPU SkyLake Gen CPUs. We had access to one Kyle node.

Partitioning across too many processes actually saw longer run times. Ideal operating point was around 6 to 10 processors. Here is an example of an MPI command used: `mpirun -n 4 python3 main.py`

4.3 Execution Time and Simulation Parameters

The execution time for the linear solver varied depending on the mesh refinement. For the coarsest mesh, the solver required approximately 35 seconds, whereas for the finest mesh, runtimes ranged from **7 to 8 minutes**.

The inflow conditions were set to a constant velocity of **500 m/s**. This led to some discontinuity at the no-slip wall boundary conditions; however, the impact on the solver and the results was negligible.

Care was taken to:

- Distribute summation of contributions from each rank for global values of the objective function, lift, and drag.
- Ensure that the lift, drag, and inflow vectors were properly rotated in accordance with the bounding box rotation.

4.4 Analysis of Results

Results for the velocity field, pressure field, and the drag and lift force are plotted below.

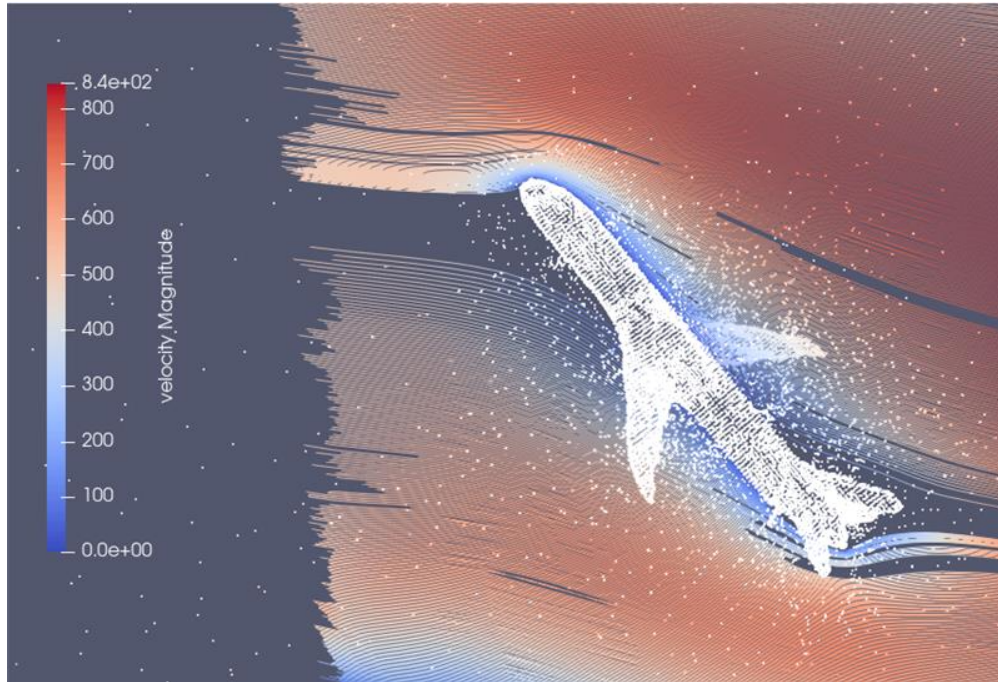


Figure 10: Velocity field visualization around the Boeing 767 model.

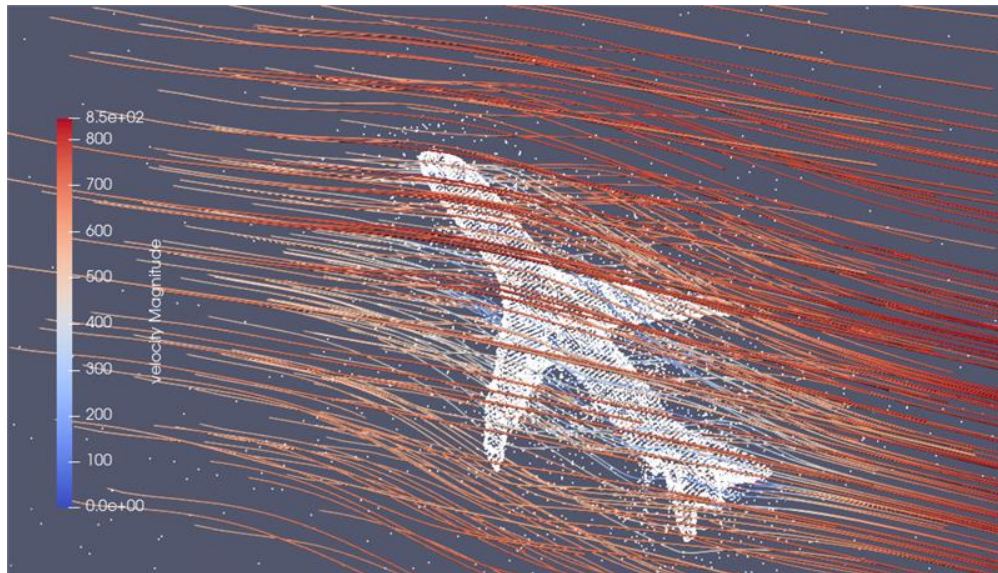


Figure 11: Zoomed-in view of the velocity field near critical regions.

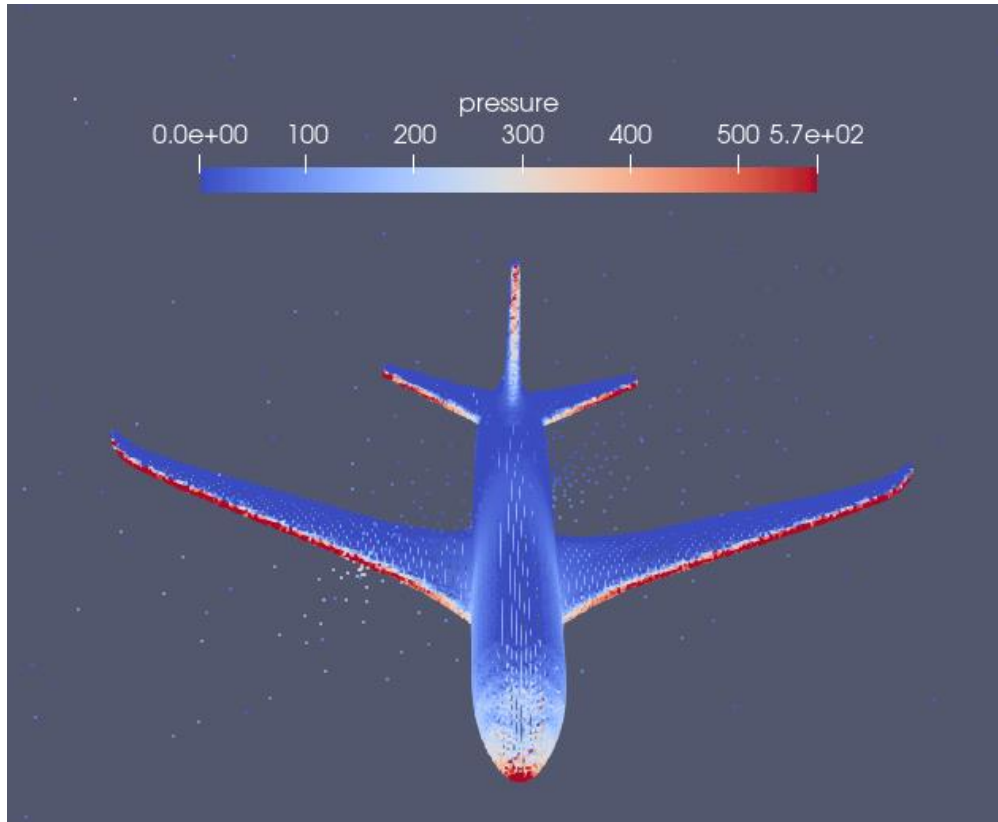


Figure 12: Pressure distribution around the Boeing 767 model.

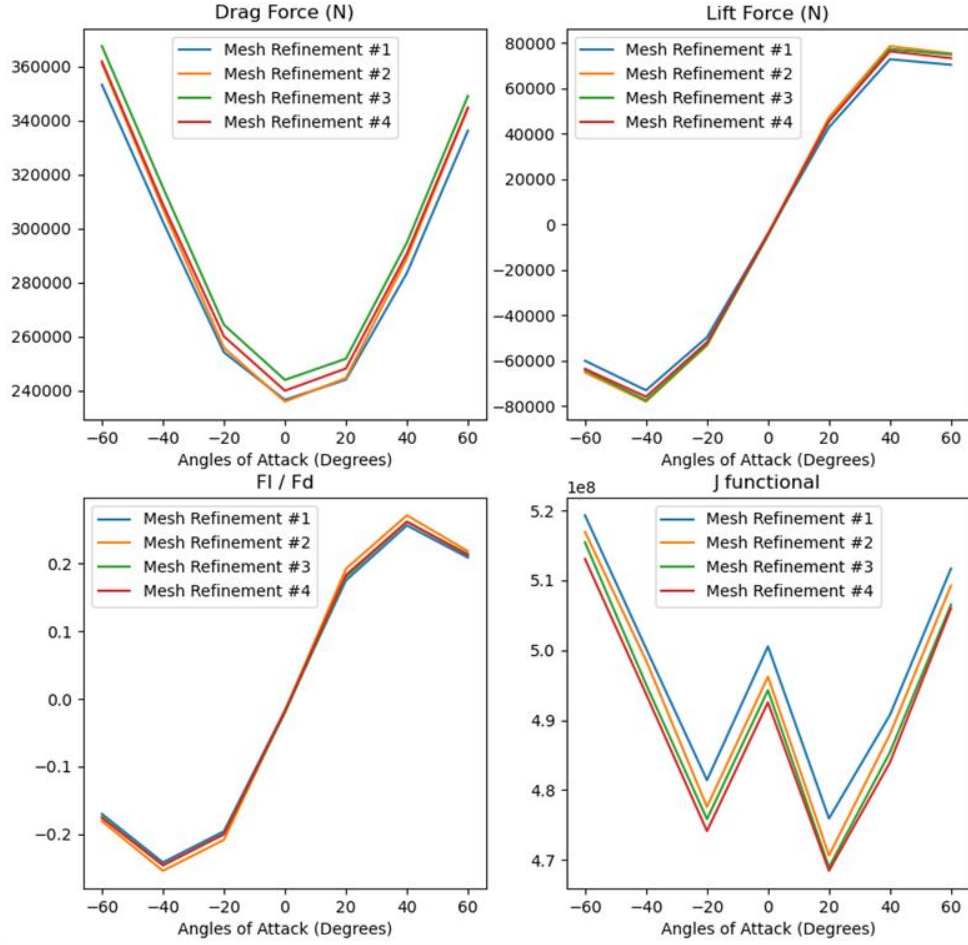


Figure 13: Lift and drag coefficient results across different mesh refinements.

The computed lift and drag values were consistent across the different levels of mesh refinement. As expected, the values obtained were relatively large, which is reflective of the size of the 3D model and the speed of the inflow. Further refinements and alternative meshing techniques, such as smoothing and quad remeshing, were tested to improve the accuracy and stability of the simulation results.

5 Simulations with Lockheed F-117 Nighthawk

Meshes were successfully created for this aircraft. However due to time constraints, no simulations were run on this model. The meshes have been attached as part of this report for the reader's reference.